

Be careful when you relabel
Constants.

$$\text{If } y = x^3 + 5C_1$$

we can relabel $5C_1$ as C .

5 times a constant is still a constant.

$$\Rightarrow y = x^3 + C$$

$$\text{If } y = x^3 + C_1 x^4$$

we can't relabel $C_1 x^4$ as
a constant.

$$\boxed{C_1 x^4 \neq C}$$

$C_1 x^4$ is not constant, it
depends on x .